

From Crowd Knowledge to AI-Assisted Development: A Longitudinal Semantic Analysis of Stack Overflow After 2020

Roham Koohestani
Delft University of Technology
Delft, The Netherlands
r.koohestani@student.tudelft.nl

Aditya Patil
Delft University of Technology
Delft, The Netherlands
a.c.patil-1@student.tudelft.nl

Tomasz Soróbka
Delft University of Technology
Delft, The Netherlands
t.j.sorobka-1@student.tudelft.nl

Abstract

Stack Overflow has served as the central knowledge-sharing platform for software developers for close to two decades. The public release of ChatGPT in November 2022 triggered a structural shift in how the platform is used. We analyse 24 million questions from the official Stack Exchange data dump (60 million posts, 189 million rows across five tables), covering 75 consecutive months from January 2020 through March 2026. Monthly question volume fell by 98.6 %, median response times tripled, and question body length and code content each grew by roughly one third; we find this to be consistent with an “AI verification tax” where developers reserve Stack Overflow for problems that AI cannot resolve. Intent classification across nine categories reveals a tenfold increase in AI-debugging questions and an 8.1-percentage-point decline in basic API-usage queries. All four technology domains (Legacy Stable, Fast-Moving Web, AI/ML, Other) lost 69–76 % of their mean monthly volume. Chow structural-break tests confirm significant breaks at the ChatGPT release for every tracked metric ($p < 0.012$), and interrupted time-series regressions quantify both level and slope changes. The full pipeline and analysis code are publicly reproducible.

Keywords

Stack Overflow, Developer Communities, Longitudinal Analysis, Generative AI, Topic Modelling, Structural Break Analysis

ACM Reference Format:

Roham Koohestani, Aditya Patil, and Tomasz Soróbka. 2026. From Crowd Knowledge to AI-Assisted Development: A Longitudinal Semantic Analysis of Stack Overflow After 2020. In *Proceedings of Web Science and Engineering (WebSciEng '26)*. ACM, New York, NY, USA, 7 pages.

1 Introduction

1.1 Motivation

Stack Overflow has served as the public memory of software engineering since its launch in 2008. Developers post technical questions, share solutions, and document issues that official manuals fail to cover. The platform’s structured data (timestamps, voting scores, tags, rich text) makes it a well-suited subject for Web Science research.

The release of ChatGPT in November 2022 [5] disrupted this model. Developers can now obtain instant answers from an AI assistant instead of posting to a public forum. Programming assistance has moved from a shared public conversation to a private one, which threatens the accumulation of new public knowledge.

Stack Overflow has not become obsolete, but the reason people use it has changed.

The 2025 Stack Overflow Developer Survey puts numbers on this change: 84 % of professional developers use or plan to use AI tools, yet 46 % distrust AI accuracy. The most cited frustration (66 %) is “AI solutions that are almost right, but not quite”, and 45 % report that debugging AI-generated code consumes excessive time.

1.2 Problem Statement

The decline in Stack Overflow activity since late 2022 is well documented. We argue that the underlying shift is an *AI verification tax*: AI handles routine coding queries privately, so the questions that reach Stack Overflow are harder, more architectural, and more frequently about debugging AI-generated logic. Stack Overflow itself recognised this change and expanded support for open-ended, opinion-based questions in early 2026. This paper measures that transition with empirical data.

1.3 Research Questions

RQ1 How have Stack Overflow activity levels, answer availability, and response speeds changed since 2020?

RQ2 Did the difficulty and complexity of Stack Overflow questions increase after 2022, and does this reflect the verification tax?

RQ3 How did the distribution of topics and developer intent shift since 2020?

RQ4 Which technical domains show the strongest changes, and does this reveal AI concept drift?

RQ5 Are the observed shifts statistically significant structural breaks, and how do level and slope changes decompose at the ChatGPT release?

1.4 Contributions

- (1) A 75-month longitudinal study of 24 million Stack Overflow questions with concrete numerical findings for five research questions.
- (2) A nine-category intent taxonomy that extends the Beyer et al. framework [3] with two AI-era categories.
- (3) A per-domain comparison across four technology groups that documents uniform decline.
- (4) Formal structural-break analysis (Chow, CUSUM, interrupted time-series regression) at the ChatGPT release date.

2 Background and Related Work

2.1 Stack Overflow as a Developer Knowledge Platform

Stack Overflow launched in 2008 and has since become the largest question-and-answer site for software developers. Questions are tagged by technology, answers are ranked by community votes, and questioners mark a single answer as accepted. This structure has made the platform a primary data source for empirical software engineering research. The Google BigQuery public dataset [6] enabled much of this work, but that snapshot was last refreshed on 2022-09-25 and cannot be used for AI-era studies.

2.2 Developer Intent and Question Taxonomies

Barua et al. [2] applied LDA topic modelling to identify discussion topics and trends on Stack Overflow. Allamanis and Sutton [1] classified questions by topic, type, and code structure. Wang et al. [7] studied developer interaction patterns. Beyer et al. [3] introduced a seven-category taxonomy of question types (API Usage, Discrepancy, Errors, Review, Conceptual, API Change, Learning) and evaluated automated classifiers. We extend the Beyer taxonomy with two AI-era categories.

2.3 Developer Behaviour in the Generative AI Era

Da Silva et al. [4] examined the reliability and impact of LLMs on Stack Overflow activity and documented early signs of the decline. Our study adds a longer observation window (through March 2026), formal structural-break tests, and domain-level decomposition.

3 Data Collection

3.1 Choice of Data Source

We initially planned to query the Stack Overflow public dataset on Google BigQuery. That snapshot was last refreshed on 2022-09-25, two months before ChatGPT's release, and is therefore useless for our purposes. A community-mirrored Stack Exchange dump on the Internet Archive proved incomplete, ending in early 2024. We therefore use the **official Stack Exchange data dump**, downloaded from the Stack Overflow Data Dump settings page. Our copy covers all posts through **2026-03-31**. The archive's SHA-256 fingerprint is recorded in `results/tables/manifest.yaml`.

3.2 Tables Used

The official dump contains eight tables. We extract five: **Posts** (questions and answers, distinguished by `PostTypeId`), **Users**, **Comments**, **PostLinks**, and **Tags**. The remaining three (**Badges**, **PostHistory**, **Votes**) are irrelevant to our research questions.

3.3 Local Ingestion Pipeline

Each table is extracted from a single .7z archive. We stream XML through `lxml.iterparse` (constant-memory parsing) and write monthly-partitioned Parquet via `pyarrow.ParquetWriter`. The partition key is `year_month=YYYY-MM`, derived from each row's `CreationDate`. The full pipeline finishes in approximately three

Table 1: Dataset summary after ingestion.

Table	Rows	Parts.	First Date	Last Date
Posts	60,349,494	213	2008-07-31	2026-03-31
Users	31,031,671	213	2008-07-31	2026-03-31
Comments	91,268,612	212	2008-08-01	2026-03-31
PostLinks	6,523,169	192	2010-04-26	2026-03-31
Tags	65,941	1	—	—
Total	189238887			

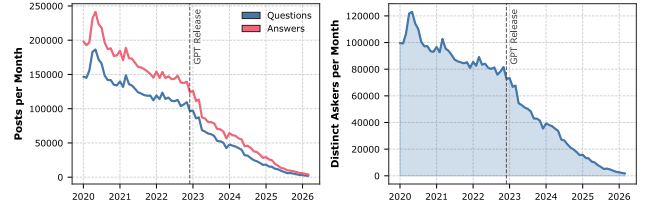


Figure 1: Monthly question and answer volume (left) and distinct askers (right). The dashed line marks the ChatGPT release.

hours on a 2024 MacBook Air; the 60-million-row `Posts` table dominates the runtime. A private Hugging Face repository hosts the materialised Parquet for co-author access.

3.4 Data Validation

Table 1 shows the ingested dataset. The analysis window covers **75 consecutive months** from January 2020 through March 2026 with **zero gaps**. Question volume dropped from 146,664 in January 2020 to 2,033 in March 2026, a 98.6% decline.

3.5 Sampling Methodology

Metrics computed via SQL aggregation (`RQ1`, `RQ4`) operate on the *full population* of six million questions in the analysis window. Analyses that require per-post text processing (`RQ2`, `RQ3`) use a power-analysis-driven stratified sample. Per month, the required sample size is the larger of two values: (1) a power-based criterion (Cohen's $h = 0.2$, $\alpha = 0.05$, power = 0.95) and (2) a margin-of-error criterion targeting ± 2 percentage points with Bonferroni correction for k simultaneous comparisons. Both apply a finite-population correction. The resulting cap is approximately 4,700 questions per month for intent classification ($k = 9$) and 4,250 for lexical features ($k = 6$), yielding 312,000–343,000 sampled posts (4.7–5.2% of the population). Sampling is deterministic via hash-ordered row numbers and thus reproducible.

4 Data Analysis

4.1 Activity Trends and Interaction Speeds (RQ1)

We track seven monthly time series across 75 months, split at the ChatGPT release on 2022-11-30 (35 pre-release months, 40 post). Figures 1 to 4 show the results.

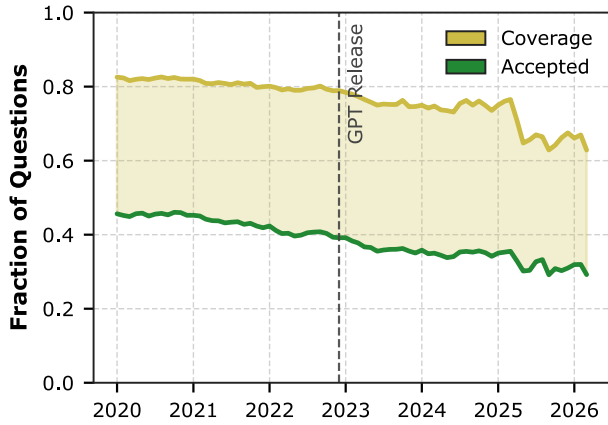


Figure 2: Monthly accepted-answer rate and answer coverage rate.

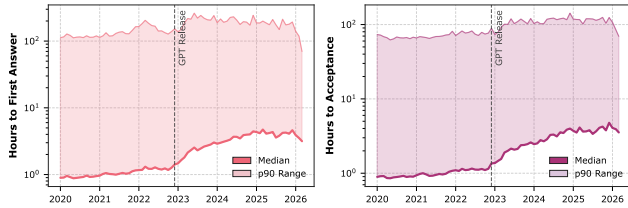


Figure 3: Median hours to first answer (left) and to acceptance (right), with p90 bands. Log scale.

Volume. Question and answer volume collapsed. Median monthly questions fell from 129,018 pre-ChatGPT to 29,682 post (-77%). The endpoint contrast is sharper: 146,664 questions in January 2020 versus 2,033 in March 2026 (-98.6%). Answers tracked questions in lockstep: median 166,903 to 43,968 per month (-74%).

Active askers. Distinct non-anonymous askers per month fell from a pre-ChatGPT median of 90,649 to 25,130 (-72%). The questions-per-asker ratio decreased from 1.42 to 1.18 (-17%). The platform thinned because *fewer people came at all*; the surviving users did not shed most of their questions.

Quality. The accepted-answer rate fell from 43.2% pre to 35.0% post (-8 pp). Answer coverage dropped from 80.9% to 73.9% (-7 pp). The gap between coverage and acceptance stayed flat (37.6% pre vs. 38.8% post), which contradicts the hypothesis that the community keeps answering while askers stop returning to accept.

Response times. Median time-to-first-answer tripled: 1.0 h pre to 3.5 h post ($\times 3.4$). Time-to-acceptance followed: 1.0 h to 2.8 h ($\times 2.9$). The median rose faster than the p90 tail, so the slowdown is platform-wide and not concentrated in hard questions.

Engagement. Mean question score fell from 0.82 to 0.61 (-25%). Mean comment count rose from 2.03 to 2.20 ($+8\%$). The surviving questions receive more discussion but fewer upvotes.

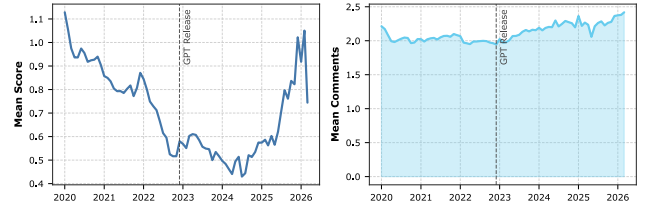


Figure 4: Mean question score (left) and mean comments per question (right).

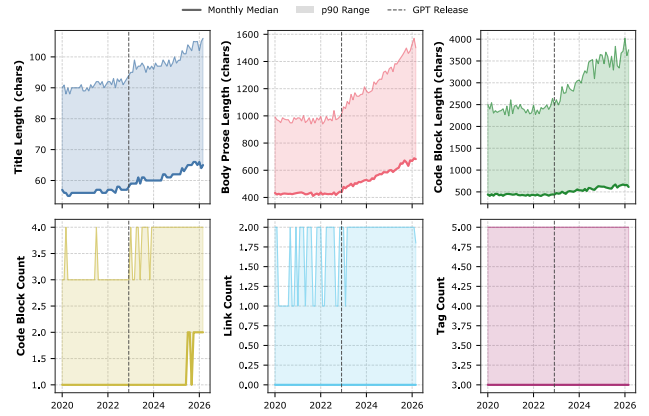


Figure 5: Six lexical features over time: monthly median (line) and p90 band. The dashed line marks the ChatGPT release.

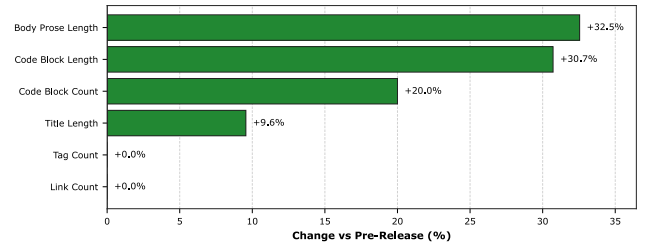


Figure 6: Percentage change in mean-of-monthly-medians, pre vs. post ChatGPT release.

4.2 Question Complexity and Difficulty (RQ2)

We compute six lexical features per question: title length, body prose length (excluding `<pre>` regions), total code-block length, code-block count, link count, and tag count. Features are extracted on a stratified monthly sample (cap 4,252 per month) and aggregated to monthly median and p90. Figure 5 plots the time series; Figure 6 shows pre/post percentage changes.

Body prose length. showed the largest change. The monthly median rose from 428 characters pre to 567 post ($+32.3\%$); the p90 rose from 972 to 1,273 ($+30.9\%$).

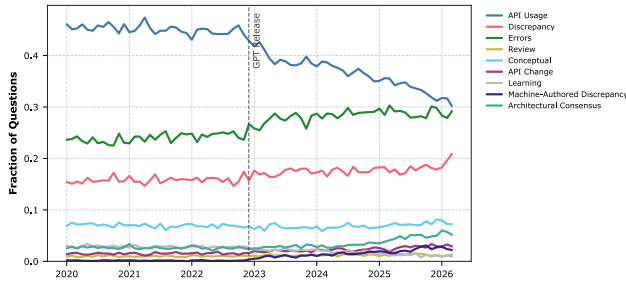


Figure 7: Monthly fraction of questions per intent category.

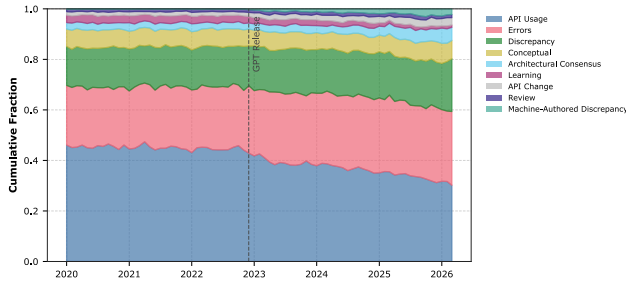


Figure 8: Stacked area chart of cumulative intent fraction.

Code-block total length. increased from 432 to 565 characters (+30.9%), mirroring prose growth. Post-release questions contain more text *and* more code; one did not substitute for the other.

Code-block count. rose at the median from 1.00 to 1.18 (+17.5%). By mid-2025 the typical question contains two separate code blocks.

Title length. grew from 56.5 to 61.8 characters (+9.3%).

Link count. held at zero at the median; the p90 rose from 1.57 to 1.98 (+25.7%).

Tag count. was flat (median 3, p90 5) because the UI enforces a hard five-tag cap.

Every content-size feature increased, while platform-constrained features held flat. Median and p90 grew within three percentage points of each other for every length feature, so the shift is population-wide. The pattern fits the verification-tax mechanism: questions that AI resolves no longer reach Stack Overflow, and the residual is heavier in every free-to-vary content dimension.

4.3 Topic and Intent Distribution (RQ3)

Each sampled question is classified into one of nine intent categories by regex-based pattern matching on the title and parsed body prose. The taxonomy extends the Beyer et al. seven-category framework [3] with two AI-era additions: *Machine-Authorized Discrepancy* (debugging AI-generated code) and *Architectural Consensus* (seeking design validation). The sample comprises 4,686 questions per month (342,945 total); patterns are evaluated in specificity order, most specific first.

Table 2: Intent category fractions pre and post ChatGPT release.

Intent	Pre (%)	Post (%)	Δ (pp)
API Usage	45.1	37.0	-8.1
Errors	24.0	28.2	4.2
Discrepancy	15.7	17.6	1.9
Conceptual	7.0	6.8	-0.2
Arch. Consensus	2.7	3.5	0.9
Learning	2.8	1.8	-1.0
API Change	1.5	2.3	0.8
Mach.-Auth. Disc.	0.15	1.5	1.4
Review	1.0	1.2	0.2

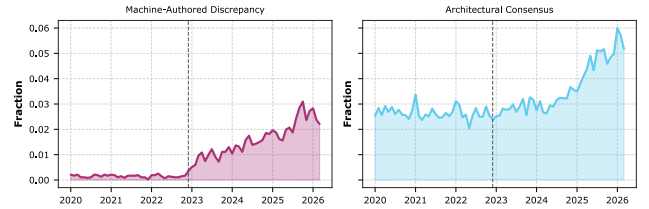


Figure 9: The two AI-era intent categories: Machine-Authorized Discrepancy (left) and Architectural Consensus (right).

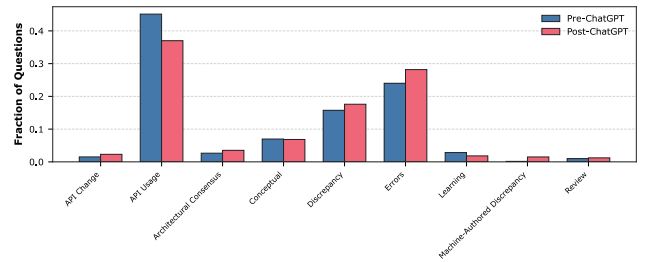


Figure 10: Pre vs. post ChatGPT intent fractions (grouped bars).

Figure 7 plots the nine category fractions over time, and Figure 8 gives a stacked-area view of the same data. Table 2 quantifies the pre/post shift.

Decline in API Usage. API Usage, the “how do I use X” category, fell from 45.1 % to 37.0 % (−8.1 pp). AI assistants handle these queries well, and the decline is consistent with displacement to private tools.

Rise in error diagnosis. Errors rose from 24.0 % to 28.2 % (+4.2 pp), and Discrepancy (unexpected behaviour) rose from 15.7 % to 17.6 % (+1.9 pp). Troubleshooting categories now account for 45.8 % of post-release questions, up from 39.7 %.

Machine-Authorized Discrepancy. This AI-era category grew ten-fold, from 0.15 % pre to 1.50 % post. The absolute share remains small, but Figure 9 shows accelerating growth from 2023 onward as AI-generated code becomes more prevalent.

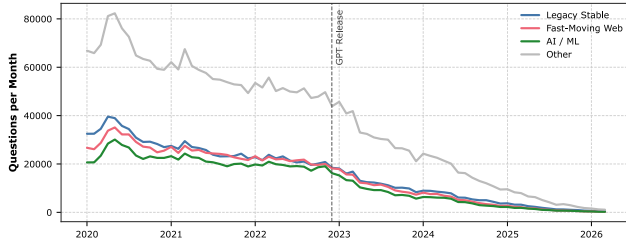


Figure 11: Monthly question volume by technology domain.

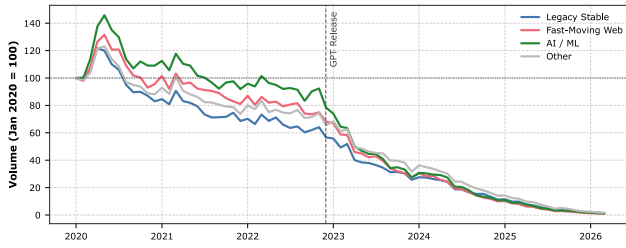


Figure 12: Volume indexed to January 2020 = 100 per domain.

Architectural Consensus. grew from 2.7 % to 3.5 % (+0.9 pp). Developers seek human judgement on design decisions that AI tools cannot resolve authoritatively.

Decline in Learning. Learning-oriented questions fell from 2.8 % to 1.8 % (−1.0 pp). Introductory material is now obtained from AI instead.

4.4 Tag-Group and Domain Comparison (RQ4)

We partition questions into four domains based on tag membership to test whether the ChatGPT-era shift is uniform or exhibits concept drift: *Legacy Stable* (C++, C#, Java, SQL, etc.), *Fast-Moving Web* (JavaScript, React, Angular, Node.js, etc.), *AI/ML* (Python, TensorFlow, PyTorch, etc.), and *Other*. All domain metrics use the full population.

Volume. Figure 11 shows absolute volume; Figure 12 normalises to January 2020. Table 3 reports the decline. All four domains lost 69–76 % of their mean monthly volume. AI/ML (−75.5 %) and Fast-Moving Web (−75.0 %) fell slightly more than Legacy Stable (−73.4 %) and Other (−69.2 %), but the differences are small. The concept-drift hypothesis, which predicts that rapidly evolving frameworks would suffer disproportionately, finds only weak support.

Quality. Figure 13 shows accepted rate and coverage rate by domain. AI/ML had the largest accepted-rate drop (46.8 % to 37.9 %, −8.9 pp); Fast-Moving Web dropped from 44.6 % to 36.0 % (−8.6 pp); Legacy Stable fell from 43.0 % to 37.0 % (−6.0 pp). Coverage rates followed the same pattern.

Response times. Figure 14 confirms that every domain experienced rising time-to-first-answer. No domain escaped the slowdown.

Table 3: Mean monthly volume change by domain, pre vs. post.

Domain	Pre mean	Post mean	Change (%)
AI / ML	21,722	5,326	−75.5
Fast-Moving Web	25,229	6,316	−75.0
Legacy Stable	26,815	7,144	−73.4
Other	59,298	18,235	−69.2

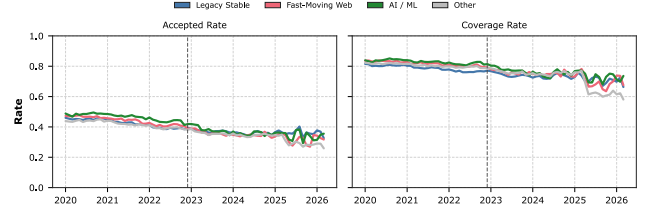


Figure 13: Accepted rate (left) and coverage rate (right) by domain.

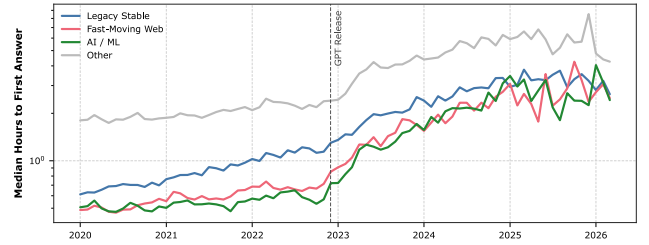


Figure 14: Median hours to first answer by domain (log scale).

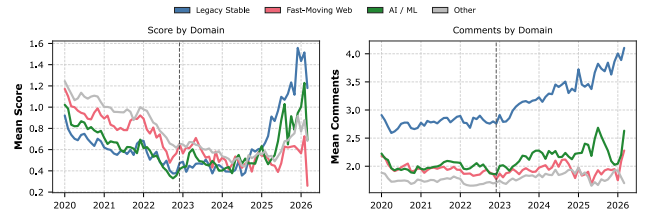


Figure 15: Mean score (left) and mean comments (right) by domain.

Engagement. Figure 15 shows mean score and comments by domain. Legacy Stable carried the highest comment counts: 2.77 pre, 3.13 post (+13 %). Established-technology questions attract more back-and-forth.

4.5 Temporal Statistics (RQ5)

We apply three formal statistical tests to each of the nine key time series, with the ChatGPT release (December 2022) as the hypothesised breakpoint.

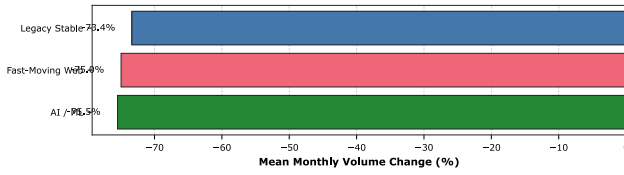
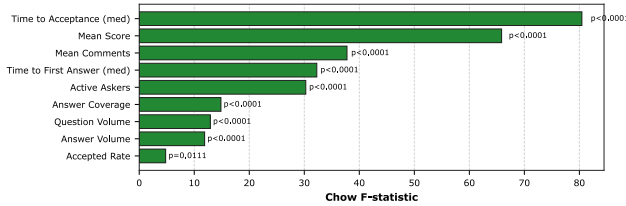


Figure 16: Mean monthly volume change (%) by domain.

Table 4: Chow structural-break test results at December 2022.

Metric	F -stat.	p -value
Time to Accept. (med)	80.46	$< 10^{-6}$
Mean Score	65.86	$< 10^{-6}$
Mean Comments	37.76	$< 10^{-6}$
Time to 1st Ans. (med)	32.28	$< 10^{-6}$
Active Askers	30.27	$< 10^{-6}$
Answer Coverage	14.85	4×10^{-6}
Question Volume	12.92	1.6×10^{-5}
Answer Volume	11.89	3.5×10^{-5}
Accepted Rate	4.80	0.011

Figure 17: Chow F -statistics per metric. Green bars: $p < 0.05$.

4.5.1 Chow Structural-Break Test. The Chow test fits a segmented linear regression (intercept, trend, post-break indicator, and interaction term) and compares its residual sum of squares against a restricted model with no break. Table 4 shows that **all nine metrics** exhibit a significant structural break at $\alpha = 0.05$. The strongest signals come from Time to Acceptance ($F = 80.46$, $p < 10^{-6}$), Mean Score ($F = 65.86$), and Mean Comments ($F = 37.76$). The weakest, Accepted Rate ($F = 4.80$, $p = 0.011$), is still significant.

4.5.2 CUSUM Test. The CUSUM test accumulates OLS residuals under a null hypothesis of linear trend and compares the maximum cumulative deviation against the 5% critical boundary. All nine metrics exceed the critical value (8.21), confirming pervasive structural instability. Volume metrics peak in March 2023, four months after the release. Response-time metrics peak later (October 2023 for first answer). The lag points to a delayed cascade from volume collapse to answering-pipeline deterioration.

4.5.3 Interrupted Time-Series Regression. We fit the ITS model $y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 T_t^{\text{post}} + \varepsilon_t$, where D_t is a post-break indicator and T_t^{post} counts months since the break. Table 5 reports the level change (β_2) and slope change (β_3).

Question volume experienced a significant level drop ($-18,960$ questions/month, $p < 10^{-4}$) and a steepening decline (-502 additional

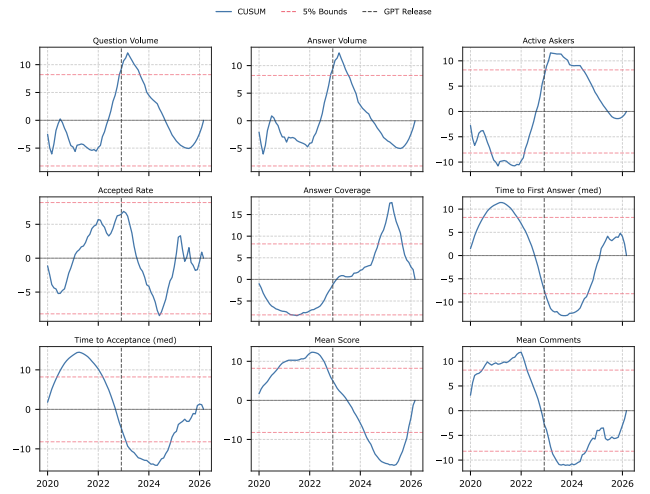


Figure 18: CUSUM paths per metric with 5% critical bounds (dashed).

Table 5: ITS regression: level and slope change at December 2022.

Metric	β_2 (level)	p	β_3 (slope)	p
Question Vol.	-18,960	$< 10^{-4}$	-502	0.010
Answer Vol.	-22,234	$< 10^{-4}$	-454	0.044
Active Askers	-10,722	$< 10^{-4}$	-808	$< 10^{-6}$
Accepted Rate	-0.014	0.004	+0.0001	0.641
Ans. Coverage	+0.003	0.713	-0.002	$< 10^{-6}$
1st Answer (s)	+2,743	$< 10^{-4}$	+194	$< 10^{-6}$
Acceptance (s)	+1,748	$< 10^{-4}$	+226	$< 10^{-6}$
Mean Score	-0.130	0.004	+0.021	$< 10^{-6}$
Mean Comments	+0.047	0.087	+0.011	$< 10^{-6}$

questions lost per month, $p = 0.01$). Response times show the most dramatic ongoing deterioration: time-to-acceptance gains +226 seconds per month ($p < 10^{-6}$), an accelerating slowdown that has not stabilised by March 2026. Mean Score dropped immediately (-0.13 , $p = 0.004$) but the positive slope change ($+0.021$ /month, $p < 10^{-6}$) indicates gradual recovery in the quality of surviving questions. Answer Coverage had no significant level change ($p = 0.713$) but a significant negative slope (-0.002 /month, $p < 10^{-6}$), indicating slow erosion rather than an abrupt drop.

All ITS models achieve R^2 values between 0.74 and 0.98; the segmented trend model captures the vast majority of variance.

5 Limitations and Threats to Validity

Data source. The BigQuery snapshot ended on 2022-09-25, which forced our pivot to the official dump. Community-mirrored dumps can omit posts from deleted users; we use the official release to minimise this gap.

Confounding platform events. Stack Overflow opened opinion-based questions to all users in early 2026 and launched a major site redesign between February and April 2026 (later withdrawn).

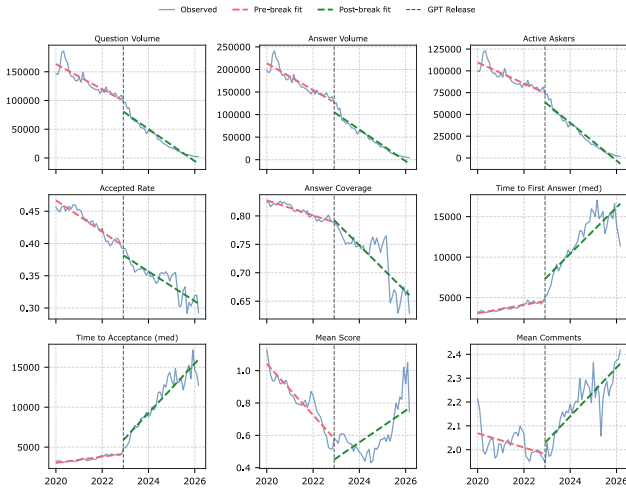


Figure 19: ITS fits per metric. Solid: observed. Dashed: fitted segments.

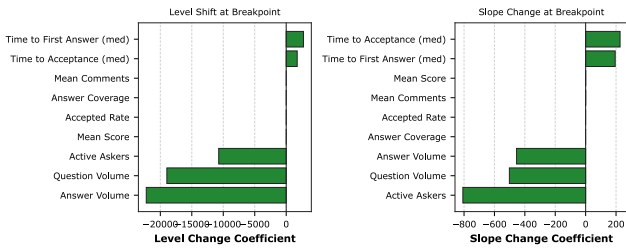


Figure 20: ITS level change (left) and slope change (right). Green = significant at $p < 0.05$.

These events affect engagement in the late period of our window. The structural-break tests are robust to late-period noise because the breakpoint is fixed at December 2022.

Selection bias. Our data covers public developer activity only. Private AI tool usage inside companies is invisible. The 98.6 % decline in public questions does not imply an equal decline in developer problems, because many of those problems migrated to private AI interactions.

Interaction metrics as proxies. Time-to-first-answer and accepted rate are practical signals of community behaviour, not direct measures of developer success.

Classification. The regex-based intent classifier operates on surface patterns. It is transparent and reproducible but will misclassify questions with atypical phrasing.

6 Conclusions

We analysed Stack Overflow from January 2020 through March 2026, addressing five research questions about the platform’s transformation in the generative AI era.

RQ1. Activity collapsed. Monthly question volume fell 98.6 %; the asker base shrank 72 %; response times tripled. Participation contraction, rather than per-user offloading, drives the decline.

RQ2. The surviving questions are heavier: body prose up 32 %, code content up 31 %, code-block count up 18 %. Every free-to-vary content dimension increased; structurally constrained features held flat.

RQ3. The intent distribution shifted away from API Usage (−8.1 pp) and toward Errors (+4.2 pp) and Discrepancy (+1.9 pp). Machine-Authorized Discrepancy grew tenfold from 0.15 % to 1.50 %.

RQ4. All four technology domains lost 69–76 % of volume. The concept-drift hypothesis finds only weak support; AI/ML and Fast-Moving Web declined marginally more than Legacy Stable.

RQ5. Chow tests confirm significant structural breaks for all nine metrics at the ChatGPT release. ITS regression reveals both level drops and slope changes; response-time deterioration accelerates at +226 seconds per month.

The evidence supports the AI verification tax hypothesis. Developers have not stopped encountering problems; they reserve Stack Overflow for the problems that AI cannot solve. The platform is transitioning from a general-purpose code reference to a specialised venue for error diagnosis, architectural consensus, and AI-code debugging.

7 Reproducibility

Researcher with their own copy of the Stack Overflow data dump clones the repository, sets DATA_DUMP to point at the dump location, and runs the pipeline. SHA-256 checksums in results/tables/manifest.yaml allow byte-level verification. All code, the dataset manifest, figure-generation scripts, and the sampling methodology are open-source in the project repository.

8 Statement of AI Use

All core ideation, conceptualisation, and research directions are the exclusive work of the authors. AI tools assisted with writing and analysis. All AI-generated suggestions were critically reviewed; the authors assume full responsibility for the final content.

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